

Single-Lagrangian CL inflow fits of the SPARC category-A galaxies: fit quality, parameter determination and robustness

Automated analysis for E.P.J. de Haas

Summary

Of the 175 SPARC galaxies, 54 fall in category A after the single-Lagrangian (single-L) constant-Lagrangian (CL) inflow fit: their weighted residuals show no significant coherent structure (runs test $p \geq 0.05$) and $\chi_\nu^2 \leq 1$. For these galaxies the two-parameter CL curve is a statistically complete description of the SPARC rotation curve. The global χ^2 is 100.7 for 341 degrees of freedom ($\chi_\nu^2 = 0.295$), the median relative RMS residual in v^2 is 0.091, and no residual exceeds 2σ . The asymptotic CL speed $v_L = \sqrt{3/2} X$ is determined to a median 5.3%, while M and R individually reach 23% and 14% because they are strongly correlated (median $\rho_{MR} = 0.967$). The fitted parameters track independent photometric quantities: v_L versus SPARC V_{flat} gives $r = 0.994$, $\log M$ versus $\log M_{\text{bar}}$ gives $r = 0.81$, and R versus R_{disk} gives Spearman $\rho = 0.68$. A three-level grading gives 17 galaxies of grade I (both parameters robustly determined), 28 of grade II (asymptotic speed determined, M and R partly degenerate) and 9 of grade III (under-constrained). The main limitation is not the model but the data: category A is dominated by late-type dwarfs with few points (median $n = 8$), and the SPARC velocity errors are conservative relative to the scatter about the CL curve, so a good fit here is partly the absence of evidence against the model.

1 Sample and selection

Category A was defined in the pipeline run on all 175 SPARC galaxies (Lelli, McGaugh & Schombert 2016): after the single-L fit, a galaxy is in A when the one-sided Wald–Wolfowitz runs test on the signs of the weighted residuals gives $p_{\text{runs}} \geq 0.05$ (no excess of long same-sign stretches), and $\chi_\nu^2 \leq 1$. Galaxies with a $+-+$ wave (B), a $-+-$ wave (C), other significant structure (D), or $\chi_\nu^2 > 1$ without structure (E) are excluded. The selection has three consequences that must be kept in mind. (i) It favours late types: 41 of 54 are Sdm–BCD (median type Sm). (ii) It favours short rotation curves: median $n = 8$, only 13 galaxies have $n \geq 10$, and 21 have fewer than two points inside the fitted R (8 have none). (iii) It includes all quality classes (18 Q=1, 29 Q=2, 7 Q=3). The runs test has low power for small n , so category A membership means *no detectable* structure, not proof that none exists.

2 Model and methods

Model. The single-L CL inflow curve (de Haas, SPARC H_z paper, Eqs. 17–18), with $X = \sqrt{2GM/R} - H_z R$:

$$v^2(r) = \frac{1}{2} X^2 r^2 / R^2 \quad (r \leq R), \quad v^2(r) = \frac{3}{2} X^2 - (\sqrt{2GM/r} - H_z r)^2 \quad (r > R),$$

with $H_z = 2.2 \times 10^{-18} \text{ s}^{-1}$ fixed, and asymptotic speed $v_L = \sqrt{3/2} X$. The fit is to v^2 with $\sigma_{v^2} = 2V_{\text{obs}} \delta V_{\text{obs}}$, by bounded least squares from 14 starting radii.

Uncertainties. Statistical: 1σ from the local covariance $(J^T J)^{-1}$; σ_{v_L} by linear propagation including the M – R covariance. Jackknife: leave-one-out refits, giving a jackknife error and the largest parameter shift in units of σ . Systematic: the data are refitted with the distance increased by δD (radii scale with D) and with the inclination increased by δi (velocities scale with $1/\sin i$, errors likewise); the two shifts are added in quadrature. SPARC velocity errors exclude both effects.

Fit-quality statistics. χ^2 , χ_ν^2 ; the lower-tail probability $P(\chi^2 \leq \chi_{\text{obs}}^2)$ for $n - 2$ degrees of freedom (small values signal overestimated errors); RMS_{rel} of $(v_{\text{obs}}^2 - v_{\text{mod}}^2)/v_{\text{obs}}^2$; runs test p_{runs} ; Durbin–Watson statistic DW (2 for uncorrelated residuals); Shapiro–Wilk normality p_{SW} ; fraction of points within 1σ ; and $\Delta \text{BIC}_{\text{off}}$, the BIC change on adding a constant offset V_0^2 (2018 RMWRSS variant).

Robustness checks. The fit is repeated with $H_z = 0$ (the 2018 RMWRSS model) to measure the sensitivity to the cosmological term.

Grading. **I:** $n \geq 8$, at least 2 points inside and 3 outside R , $\sigma_M/M \leq 0.3$, $\sigma_R/R \leq 0.3$, and no jackknife shift above 2σ . **II:** not I, but $\sigma_{v_L}/v_L \leq 0.10$. **III:** the rest. Galaxies with $P(\chi^2 \leq \chi_{\text{obs}}^2) < 0.01$ are flagged as having conservative errors.

3 Global fit quality

Table 1: Category-A fit-quality statistics (54 galaxies).

Statistic	16%	median	84%	range
n	6	8	11	4–25
χ_ν^2	0.039	0.177	0.656	0.004–0.930
$P(\chi^2 \leq \chi_{\text{obs}}^2)$	0.000	0.020	0.320	0.000–0.555
RMS_{rel}	0.041	0.091	0.194	0.013–0.308
p_{runs}	0.089	0.202	0.582	0.051–0.960
Durbin–Watson	0.83	1.36	2.03	0.36–2.86
p_{sw}	0.183	0.517	0.785	0.002–1.000
fraction $ \text{WR} \leq 1$	0.88	1.00	1.00	0.67–1.00
max $ \text{WR} $	0.31	0.68	1.23	0.08–1.66
ρ_{MR}	0.956	0.967	0.982	0.926–1.000
σ_M/M	0.112	0.233	0.506	0.064–1.304
σ_R/R	0.074	0.139	0.317	0.043–1.265
σ_{v_L}/v_L	0.021	0.053	0.102	0.011–0.383
$\Delta\text{BIC}_{\text{off}}$	0.33	1.60	2.06	−3.14–2.91

Residual distribution. The 449 pooled weighted residuals (Fig. 1) are centred on zero and close to Gaussian in shape, but their dispersion is only 0.47, against 1 expected for correctly scaled errors. After rescaling each galaxy by $\sqrt{\chi_\nu^2}$ the Q–Q plot is linear over $\pm 2.5\sigma$. Shapiro–Wilk rejects normality ($p < 0.05$) in only 3 galaxies (CamB, F571-V1, UGC 11557), about the 2.7 expected by chance. No residual exceeds 2σ , and only one galaxy has fewer than 68% of its points within 1σ .

Error calibration. The lower-tail probability $P(\chi^2 \leq \chi_{\text{obs}}^2)$ is far from uniform (Fig. 1, right): 32 galaxies lie below 0.05 and 21 below 0.01. With only two free parameters this cannot be overfitting; it means that SPARC’s δV_{obs} , which by construction include kinematic asymmetries and non-circular motions, are on average about 1.8 times larger than the scatter of the data about the CL curve. Two consequences follow: $\chi_\nu^2 < 1$ cannot be used to rank galaxies within category A, and the covariance uncertainties quoted below are conservative.

Residual structure. By selection $p_{\text{runs}} \geq 0.05$ throughout, but 9 galaxies lie between 0.05 and 0.1 and 18 have $\text{DW} < 1$. These are the members closest to category B or C and the first candidates for a second Lagrangian if more data become available.

Offset and H_z . No galaxy requires the constant offset: the largest improvement is $\Delta\text{BIC}_{\text{off}} = -3.1$ (UGC 6917), and the median is +1.6, so the extra parameter is penalised. Setting $H_z = 0$ changes M by at most 1.8% and R by at most 1.3%. At galactic scales the Hubble term is therefore negligible for the fit, and the 2018 RMWRSS ($H_z = 0$) and present fits are equivalent for category A.

4 Parameter determination

The M – R covariance dominates the parameter errors. The correlation coefficient ranges over 0.926–1.000. The data constrain the combination GM/R , which sets the plateau $v_L^2 \simeq 3GM/R$, much better than M and R separately, so solutions slide along lines of constant v_L (Fig. 2, left). Accordingly v_L is the best-determined quantity (median 5.3%), followed by R (14%) and M (23%). R is fixed mainly by where the inner parabola turns over into the outer branch. In the 8 galaxies with no point inside R , it is constrained only by the curvature of the outer branch $-2GM/r$, which works but is less precise.

The jackknife errors are smaller than the covariance errors (median ratio 0.61 for M), consistent with the conservative SPARC errors. The largest single-point influence exceeds 2σ in 4 galaxies. The systematic

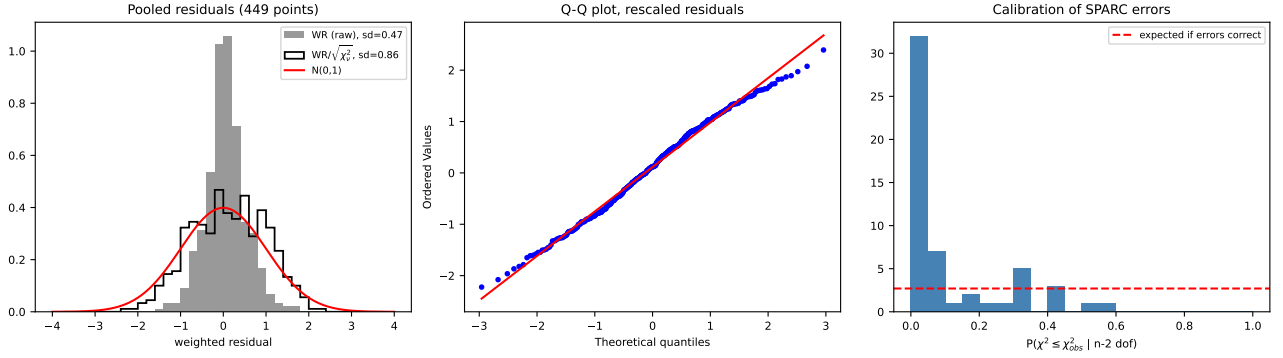


Figure 1: Left: pooled weighted residuals, raw (grey) and rescaled by $\sqrt{\chi^2_\nu}$ per galaxy (black), against $N(0,1)$. Middle: Q-Q plot of the rescaled residuals. Right: distribution of $P(\chi^2 \leq \chi^2_{\text{obs}})$; a uniform distribution (dashed) is expected if the SPARC errors were correctly scaled.

uncertainty from distance and inclination is comparable to the statistical one (median ratio 0.97 for M and 1.14 for R), so it should always be quoted alongside it. It matters most for low-inclination galaxies, through the $1/\sin^2 i$ scaling of v^2 , and for Hubble-flow distances.

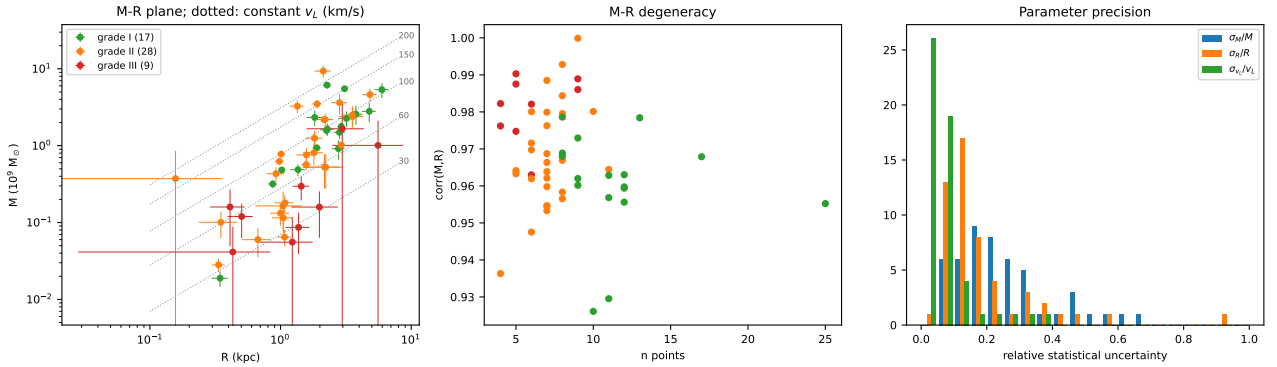


Figure 2: Left: fitted (R, M) with 1σ errors, coloured by grade; dotted lines are curves of constant v_L , along which the M - R degeneracy runs. Middle: M - R correlation versus number of points. Right: relative statistical uncertainties of M , R and v_L .

5 Physical consistency

For the 28 galaxies with a SPARC V_{flat} , the CL asymptotic speed tracks it closely ($r = 0.994$), with $v_L/V_{\text{flat}} = 1.095$ (median). The roughly 10% excess is expected rather than a bias: at the last measured radius the CL curve has reached only 0.92 of v_L (median), whereas V_{flat} is measured from the outermost points. The CL curve therefore predicts a continued slow rise of a few per cent beyond the data, which is a testable prediction for deeper HI observations. The fitted mass correlates with the SPARC baryonic mass ($M_{\text{bar}} = 0.5L_{\text{disk}} + 0.7L_{\text{bulge}} + 1.33M_{\text{HI}}$; $r = 0.81$ in log), with median $M/M_{\text{bar}} = 0.41$ (16–84%: 0.15–0.99). This is much tighter than the $r \approx 0.28$ found for the full 175-galaxy sample, and consistent with reading M as the inner (bulge-like) mass rather than the total baryonic mass. R is of the order of the disk scale length (median $R/R_{\text{disk}} = 1.11$, Spearman $\rho = 0.68$).

6 Grades and assessment

Grade I (17 galaxies, of which 11 are $Q=1$) is the core result. The rotation curve samples both the inner parabola and the outer branch, both parameters are determined to better than 30%, and no single point drives the fit. These are clean, quantitative confirmations that a single constant Lagrangian describes the full rotation curve. **Grade II** (28) fits as well statistically, but the data constrain mainly v_L . M and R are partly degenerate, usually because of few inner points. These galaxies support the model’s plateau and shape but give weaker individual scale parameters. **Grade III** (9) are consistent with the model but not informative: with $n \leq 9$ points and parameter errors up to 122%, almost any smooth rising curve would fit.

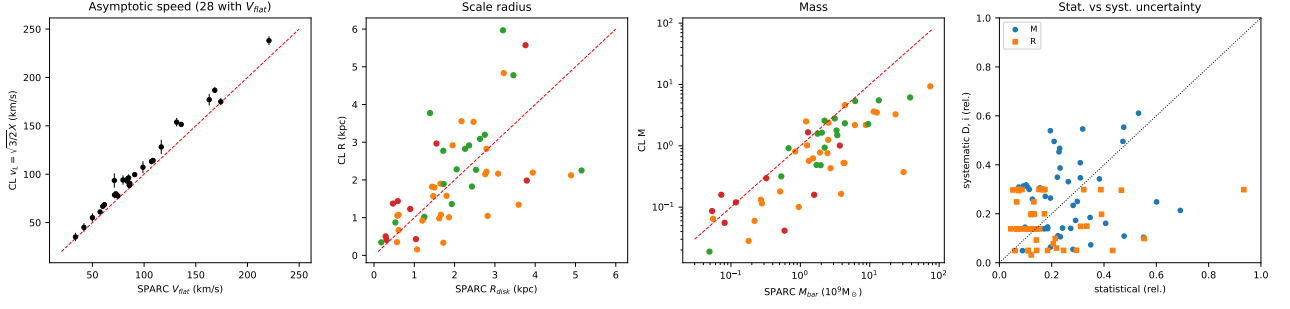


Figure 3: Left to right: CL v_L versus SPARC V_{flat} ; R versus R_{disk} ; M versus M_{bar} (dashed: one-to-one); systematic (distance, inclination) versus statistical relative uncertainty of M and R .

Table 2: Grades (medians per grade).

Grade	N	n	χ^2_ν	RMS_{rel}	σ_M/M	σ_R/R	σ_{v_L}/v_L
I	17	11	0.220	0.126	0.175	0.111	0.038
II	28	7	0.156	0.068	0.237	0.153	0.045
III	9	5	0.082	0.143	0.692	0.390	0.174

Overall assessment. Within category A the single-L CL model is statistically sufficient. It has no significant residual structure, Gaussian-shaped residuals, no outliers, no need for an offset, and parameters that agree with independent photometric and kinematic scales. Two limitations are intrinsic to the data rather than to the model. First, the SPARC errors are conservative, so goodness of fit cannot discriminate among the A galaxies or between the CL model and other two-parameter models on χ^2 alone. Second, short curves and the M – R degeneracy mean that only about a third of the sample (grade I) determines both CL parameters robustly. For comparisons with MOND or dark-matter halos, the grade-I subset with total (statistical plus systematic) uncertainties is the appropriate sample.

Table 3: Category-A parameters. R in kpc, M in $10^9 M_\odot$, v_L in km s^{-1} . Uncertainties: statistical (covariance) $\pm$ systematic (distance and inclination); jk = jackknife error. $n_{\text{in}}/n_{\text{out}}$ = points inside/outside R .

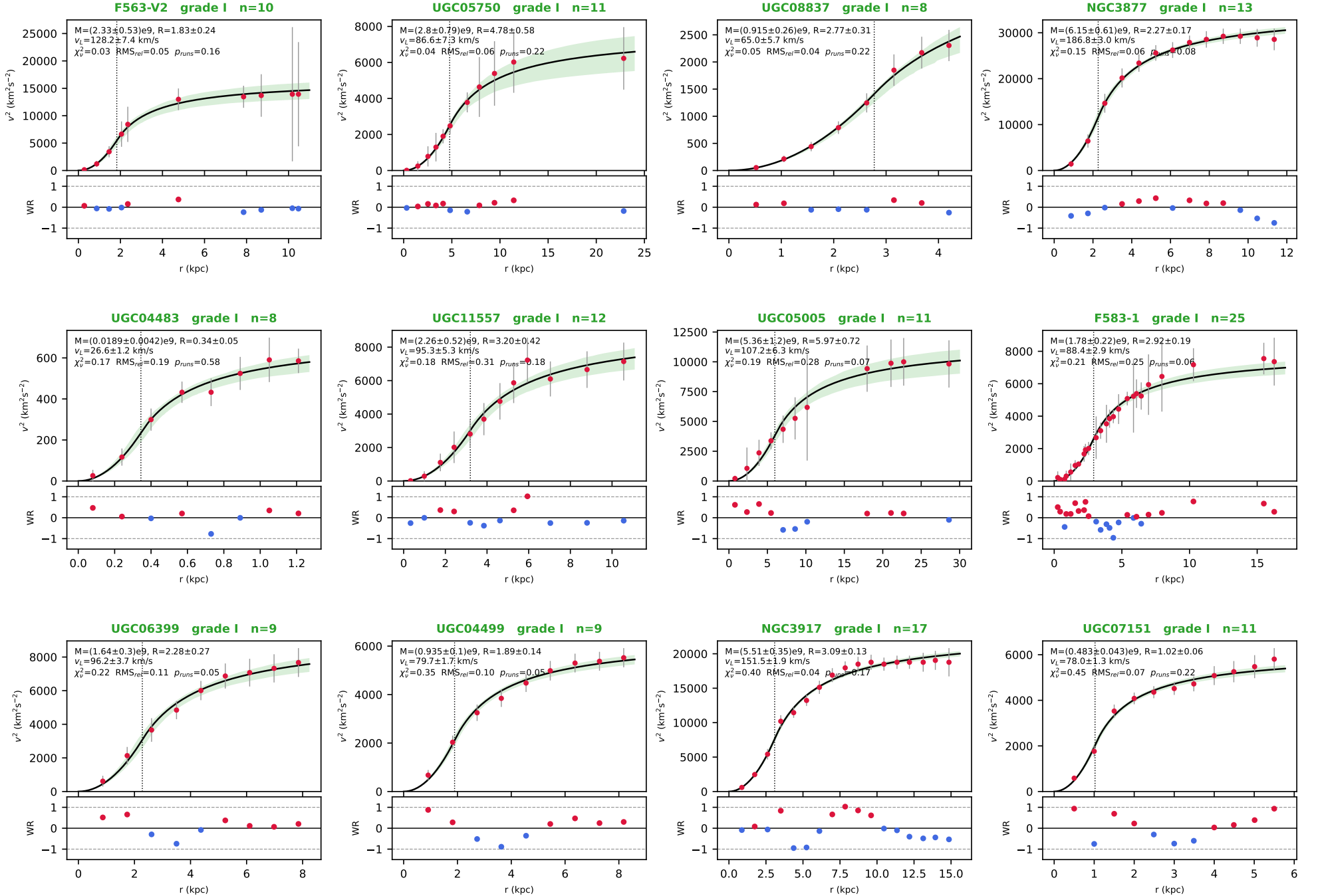
Galaxy	Type	Q	i	n	$n_{\text{in}}/n_{\text{out}}$	$R \pm \text{stat} \pm \text{sys}$	$M \pm \text{stat} \pm \text{sys}$	jk(R, M)	v_L	ρ_{MR}	V_{flat}	Grade
F563-V2	Im	1	29	10	3/7	$1.825 \pm 0.244 \pm 0.363$	$2.33 \pm 0.53 \pm 1.1$	0.052, 0.17	128.2 ± 7.4	0.936	116.6	I
F583-1	Sm	1	63	25	10/15	$2.915 \pm 0.192 \pm 0.725$	$1.78 \pm 0.22 \pm 0.46$	0.14, 0.16	88.4 ± 2.9	0.955	85.8	I
F583-4	Sc	1	55	12	3/9	$1.360 \pm 0.175 \pm 0.272$	$0.487 \pm 0.085 \pm 0.13$	0.32, 0.15	67.9 ± 2.1	0.960	—	I
NGC3877	Sc	2	76	13	2/11	$2.268 \pm 0.166 \pm 0.314$	$6.15 \pm 0.61 \pm 0.85$	0.054, 0.21	186.8 ± 3.0	0.978	168.4	I
NGC3917	Scd	1	79	17	3/14	$3.088 \pm 0.134 \pm 0.427$	$5.51 \pm 0.35 \pm 0.76$	0.092, 0.22	151.5 ± 1.9	0.968	135.9	I
UGC04483	Im	2	58	8	2/6	$0.344 \pm 0.049 \pm 0.032$	$0.0189 \pm 0.0042 \pm 0.0021$	0.0096, 0.0011	26.6 ± 1.2	0.968	—	I
UGC04499	Sdm	1	50	9	2/7	$1.893 \pm 0.137 \pm 0.565$	$0.935 \pm 0.1 \pm 0.29$	0.14, 0.098	79.7 ± 1.7	0.962	72.8	I
UGC05005	Im	1	41	11	4/7	$5.968 \pm 0.723 \pm 1.178$	$5.36 \pm 1.2 \pm 1.9$	0.51, 0.67	107.2 ± 6.3	0.930	98.9	I
UGC05750	Sdm	1	64	11	5/6	$4.775 \pm 0.582 \pm 0.945$	$2.8 \pm 0.79 \pm 0.66$	0.17, 0.24	86.6 ± 7.3	0.957	—	I
UGC06399	Sm	1	75	9	2/7	$2.280 \pm 0.265 \pm 0.316$	$1.64 \pm 0.3 \pm 0.23$	0.21, 0.22	96.2 ± 3.7	0.973	85.0	I
UGC06667	Scd	1	89	9	2/7	$2.251 \pm 0.152 \pm 0.311$	$1.58 \pm 0.17 \pm 0.22$	0.3, 0.31	94.9 ± 2.3	0.960	83.8	I
UGC06818	Sm	2	75	8	4/4	$3.774 \pm 0.451 \pm 0.522$	$2.58 \pm 0.7 \pm 0.36$	0.29, 0.43	93.5 ± 7.5	0.969	71.2	I
UGC07089	Sdm	2	80	12	3/9	$2.826 \pm 0.272 \pm 0.391$	$1.49 \pm 0.25 \pm 0.21$	0.38, 0.33	82.2 ± 3.4	0.959	—	I
UGC07151	Scd	1	90	11	2/9	$1.021 \pm 0.061 \pm 0.050$	$0.483 \pm 0.043 \pm 0.024$	0.088, 0.055	78.0 ± 1.3	0.963	73.5	I
UGC07603	Sd	1	78	12	2/10	$0.871 \pm 0.067 \pm 0.261$	$0.317 \pm 0.036 \pm 0.095$	0.083, 0.039	68.5 ± 1.5	0.963	61.6	I
UGC08837	Im	2	80	8	5/3	$2.773 \pm 0.307 \pm 0.138$	$0.915 \pm 0.26 \pm 0.05$	0.13, 0.13	65.0 ± 5.7	0.979	—	I
UGC11557	Sdm	2	30	12	5/7	$3.198 \pm 0.421 \pm 0.795$	$2.26 \pm 0.52 \pm 1.1$	0.17, 0.21	95.3 ± 5.3	0.956	—	I
D512-2	Im	2	56	4	1/3	$0.997 \pm 0.165 \pm 0.298$	$0.132 \pm 0.041 \pm 0.046$	0.076, 0.015	41.3 ± 3.4	0.936	—	II
D564-8	Im	2	63	6	2/4	$1.074 \pm 0.131 \pm 0.034$	$0.0647 \pm 0.015 \pm 0.0069$	0.065, 0.0066	27.8 ± 1.7	0.948	—	II
DDO170	Im	2	66	8	1/7	$2.918 \pm 0.151 \pm 0.868$	$1.02 \pm 0.075 \pm 0.31$	0.21, 0.098	66.8 ± 0.9	0.980	60.0	II
F561-1	Sm	3	24	6	1/5	$2.217 \pm 0.741 \pm 0.331$	$0.524 \pm 0.25 \pm 0.26$	0.054, 0.022	55.1 ± 4.4	0.980	50.0	II
F565-V2	Im	2	60	7	2/5	$3.557 \pm 0.611 \pm 0.711$	$2.5 \pm 0.74 \pm 0.63$	0.12, 0.17	94.9 ± 6.5	0.969	—	II
F567-2	Sm	3	20	5	1/4	$2.167 \pm 0.673 \pm 0.322$	$0.527 \pm 0.25 \pm 0.29$	0.95, 0.27	55.8 ± 5.4	0.964	—	II
F571-V1	Sd	2	30	7	1/6	$3.540 \pm 0.756 \pm 0.351$	$2.38 \pm 0.74 \pm 0.97$	0.15, 0.15	92.9 ± 5.1	0.976	83.6	II
NGC3949	Sbc	2	55	7	0/7	$1.343 \pm 0.166 \pm 0.186$	$3.27 \pm 0.6 \pm 0.48$	0.072, 0.23	177.0 ± 6.0	0.980	163.0	II
NGC3953	Sbc	1	62	8	0/8	$2.122 \pm 0.302 \pm 0.294$	$9.34 \pm 1.6 \pm 1.3$	0.34, 1.8	238.1 ± 4.5	0.993	220.8	II
NGC4085	Sc	2	82	7	2/5	$1.901 \pm 0.154 \pm 0.264$	$3.49 \pm 0.46 \pm 0.48$	0.1, 0.27	153.8 ± 4.4	0.962	131.5	II
NGC4389	Sbc	3	50	6	3/3	$2.828 \pm 0.387 \pm 0.392$	$3.63 \pm 1.1 \pm 0.63$	0.61, 1.5	128.4 ± 10.4	0.970	—	II
UGC02259	Sdm	2	41	8	0/8	$0.980 \pm 0.073 \pm 0.289$	$0.625 \pm 0.057 \pm 0.2$	0.19, 0.15	90.6 ± 1.0	0.984	86.2	II
UGC04325	Sm	1	41	8	1/7	$1.010 \pm 0.074 \pm 0.302$	$0.778 \pm 0.08 \pm 0.25$	0.078, 0.077	99.6 ± 1.8	0.967	90.9	II
UGC05414	Im	1	55	6	2/4	$1.570 \pm 0.135 \pm 0.469$	$0.567 \pm 0.088 \pm 0.17$	0.13, 0.09	68.1 ± 2.6	0.962	—	II
UGC05918	Im	2	46	8	1/7	$1.079 \pm 0.182 \pm 0.323$	$0.18 \pm 0.047 \pm 0.06$	0.073, 0.019	46.3 ± 2.6	0.957	—	II
UGC05999	Im	2	22	5	1/4	$4.835 \pm 0.594 \pm 0.957$	$4.62 \pm 0.9 \pm 2.5$	0.088, 0.14	110.6 ± 4.6	0.963	—	II
UGC06628	Sm	2	20	7	0/7	$1.046 \pm 0.406 \pm 0.312$	$1.65 \pm 0.088 \pm 0.1$	0.17, 0.032	45.0 ± 4.5	0.954	41.8	II
UGC06917	Sm	1	56	11	1/10	$2.152 \pm 0.142 \pm 0.298$	$2.18 \pm 0.21 \pm 0.32$	0.36, 0.5	114.2 ± 2.2	0.964	108.7	II
UGC06923	Im	2	65	6	1/5	$1.819 \pm 0.269 \pm 0.252$	$1.25 \pm 0.3 \pm 0.18$	0.58, 0.51	94.0 ± 4.9	0.972	79.6	II
UGC06930	Sd	1	32	10	1/9	$2.194 \pm 0.337 \pm 0.303$	$2.19 \pm 0.43 \pm 0.58$	0.25, 0.31	113.2 ± 3.0	0.980	107.2	II
UGC06973	Sab	3	71	9	0/9	$0.157 \pm 0.199 \pm 0.022$	$0.373 \pm 0.49 \pm 0.053$	0.086, 0.21	175.0 ± 3.7	1.000	174.2	II
UGC07261	Sdm	2	30	7	0/7	$0.922 \pm 0.140 \pm 0.276$	$0.431 \pm 0.09 \pm 0.21$	0.17, 0.1	77.6 ± 3.1	0.953	74.7	II
UGC07559	Im	2	61	7	2/5	$1.051 \pm 0.193 \pm 0.053$	$0.115 \pm 0.04 \pm 0.0084$	0.097, 0.016	37.5 ± 3.4	0.960	—	II
UGC07608	Im	1	25	8	3/5	$1.801 \pm 0.313 \pm 0.538$	$0.809 \pm 0.26 \pm 0.44$	0.19, 0.16	76.0 ± 6.1	0.958	—	II
UGC07690	Im	2	41	7	0/7	$0.349 \pm 0.113 \pm 0.104$	$0.101 \pm 0.039 \pm 0.034$	0.29, 0.086	61.0 ± 2.4	0.989	57.4	II
UGC07866	Im	2	44	7	2/5	$0.671 \pm 0.165 \pm 0.034$	$0.0598 \pm 0.024 \pm 0.0096$	0.098, 0.015	33.9 ± 3.2	0.955	—	II
UGC10310	Sm	1	34	7	1/6	$1.578 \pm 0.253 \pm 0.476$	$0.758 \pm 0.18 \pm 0.29$	0.16, 0.098	78.6 ± 3.4	0.966	71.4	II
UGC A281	BCD	3	67	7	2/5	$0.335 \pm 0.037 \pm 0.017$	$0.0281 \pm 0.0055 \pm 0.0018$	0.037, 0.0047	32.9 ± 1.5	0.964	—	II
CamB	Im	2	65	9	7/2	$1.373 \pm 0.283 \pm 0.106$	$0.0865 \pm 0.048 \pm 0.009$	0.2, 0.031	28.4 ± 4.9	0.989	—	III
F563-V1	Im	3	60	6	1/5	$1.983 \pm 0.774 \pm 0.393$	$0.159 \pm 0.096 \pm 0.04$	0.21, 0.035	32.0 ± 3.9	0.963	—	III
F574-2	Sm	3	30	5	2/3	$5.572 \pm 3.082 \pm 0.550$	$1.01 \pm 1.1 \pm 0.41$	0.97, 0.33	47.9 ± 13.7	0.975	—	III
NGC4068	Im	2	44	6	3/3	$1.440 \pm 0.206 \pm 0.072$	$0.297 \pm 0.1 \pm 0.055$	0.14, 0.088	51.5 ± 5.4	0.982	—	III
NGC6789	BCD	2	43	4	2/2	$0.410 \pm 0.121 \pm 0.021$	$0.159 \pm 0.11 \pm 0.034$	0.17, 0.15	70.7 ± 14.4	0.982	—	III
UGC02023	Im	2	19	5	3/2	$2.966 \pm 1.383 \pm 0.879$	$1.66 \pm 2 \pm 1$	0.67, 0.81	84.7 ± 32.4	0.988	—	III
UGC07232	Im	2	59	4	2/2	$0.504 \pm 0.110 \pm 0.030$	$0.12 \pm 0.057 \pm 0.013$	0.14, 0.077	55.3 ± 7.4	0.976	—	III
UGC07577	Im	2	63	9	6/3	$1.230 \pm 0.533 \pm 0.061$	$0.0556 \pm 0.061 \pm 0.0039$	0.24, 0.027	24.0 ± 8.1	0.986	—	III
UGC09992	Im	2	30	5	0/5	$0.431 \pm 0.403 \pm 0.129$	$0.0414 \pm 0.047 \pm 0.02$	0.099, 0.011	35.1 ± 4.3	0.990	33.6	III

Table 4: Category-A goodness of fit. $P_{<} = P(\chi^2 \leq \chi^2_{\text{obs}})$; DW = Durbin–Watson; $f_{1\sigma}$ = fraction of points with $|WR| \leq 1$; $j_{k_{\text{max}}}$ = largest leave-one-out shift of M or R in σ ; $\Delta\text{BIC}_{\text{off}}$ for an added constant offset; c = conservative-error flag ($P_{<} < 0.01$).

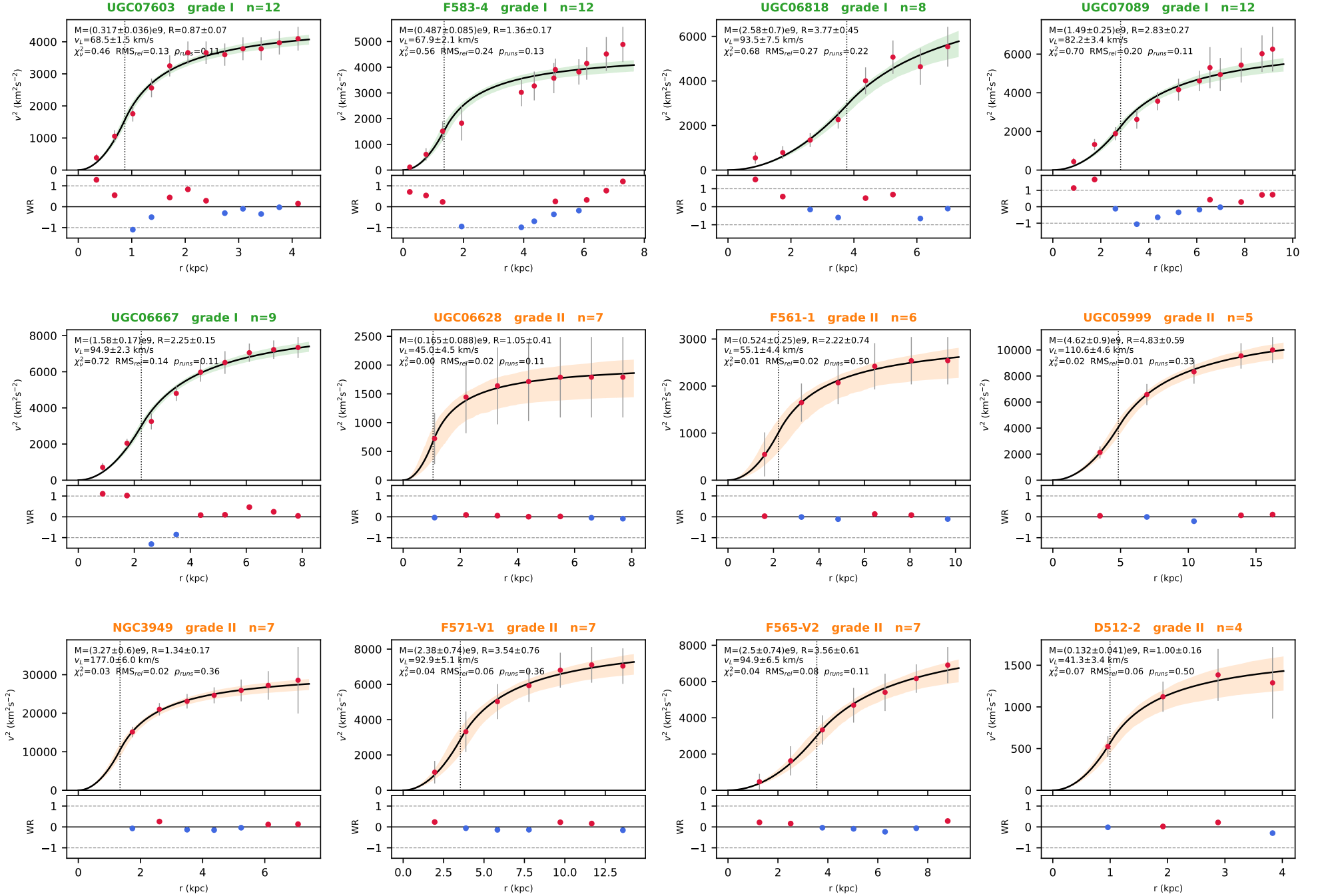
Galaxy	n	χ^2	χ^2_{ν}	$P_{<}$	AIC	BIC	RMS _{rel}	p_{runs}	DW	p_{SW}	$f_{1\sigma}$	$j_{k_{\text{max}}}$	$\Delta\text{BIC}_{\text{off}}$ / flag
F563-V2	10	0.26	0.032	0.000	4.26	4.86	0.048	0.163	1.90	0.225	1.00	0.26	+2.30 c
F583-1	25	4.90	0.213	0.000	8.90	11.33	0.250	0.059	0.82	0.605	1.00	0.40	+2.91 c
F583-4	12	5.59	0.559	0.152	9.59	10.56	0.239	0.126	0.52	0.658	0.92	1.89	+1.70
NGC3877	13	1.60	0.146	0.001	5.60	6.73	0.062	0.076	0.53	0.528	1.00	0.25	+2.30 c
NGC3917	17	5.97	0.398	0.020	9.97	11.64	0.040	0.165	1.01	0.272	0.94	0.55	+2.83
UGC04483	8	1.03	0.171	0.015	5.03	5.19	0.188	0.582	1.87	0.079	1.00	0.20	+1.85
UGC04499	9	2.44	0.348	0.068	6.44	6.83	0.100	0.051	0.76	0.506	1.00	1.06	+1.16
UGC05005	11	1.74	0.194	0.005	5.74	6.54	0.278	0.074	0.85	0.268	1.00	0.65	+1.84 c
UGC05750	11	0.33	0.036	0.000	4.33	5.12	0.065	0.225	1.60	0.734	1.00	0.17	+2.40 c
UGC06399	9	1.54	0.220	0.019	5.54	5.93	0.111	0.051	1.21	0.843	1.00	0.66	+1.56
UGC06667	9	5.01	0.716	0.342	9.01	9.41	0.145	0.110	1.35	0.410	0.67	1.84	-1.08
UGC06818	8	4.10	0.684	0.337	8.10	8.26	0.274	0.223	1.19	0.582	0.88	0.58	+0.04
UGC07089	12	7.05	0.705	0.279	11.05	12.02	0.199	0.113	0.77	0.999	0.75	1.14	-1.64
UGC07151	11	4.01	0.445	0.089	8.01	8.80	0.070	0.225	1.59	0.338	1.00	1.41	+1.37
UGC07603	12	4.64	0.464	0.086	8.64	9.61	0.126	0.113	1.20	1.000	0.83	1.01	-0.01
UGC08837	8	0.31	0.052	0.001	4.31	4.47	0.041	0.223	1.75	0.499	1.00	0.51	+2.05 c
UGC11557	12	1.83	0.183	0.003	5.83	6.80	0.308	0.179	1.59	0.020	0.92	0.31	+2.44 c
D512-2	4	0.13	0.067	0.065	4.13	2.91	0.055	0.500	2.22	0.783	1.00	0.59	+1.39
D564-8	6	0.62	0.155	0.039	4.62	4.20	0.092	0.181	1.17	0.560	1.00	0.48	+1.38
DDO170	8	2.99	0.498	0.190	6.99	7.15	0.076	0.937	2.60	0.151	1.00	1.41	+1.10
F561-1	6	0.05	0.012	0.000	4.05	3.63	0.021	0.500	2.23	0.557	1.00	0.07	+1.79 c
F565-V2	7	0.22	0.044	0.001	4.22	4.11	0.085	0.113	0.97	0.597	1.00	0.22	+1.87 c
F567-2	5	0.44	0.145	0.067	4.44	3.65	0.089	0.744	2.63	0.217	1.00	1.70	+1.57
F571-V1	7	0.20	0.039	0.001	4.20	4.09	0.060	0.358	1.66	0.046	1.00	0.14	+1.87 c
NGC3949	7	0.14	0.029	0.000	4.14	4.04	0.019	0.358	2.07	0.428	1.00	0.48	+1.95 c
NGC3953	8	3.14	0.524	0.209	7.14	7.30	0.048	0.075	0.66	0.613	0.88	1.19	+2.08
NGC4085	7	0.67	0.134	0.015	4.67	4.56	0.038	0.560	1.96	0.158	1.00	0.70	+1.91
NGC4389	6	2.98	0.745	0.439	6.98	6.56	0.176	0.181	1.17	0.241	1.00	1.09	+0.55
UGC02259	8	3.94	0.657	0.315	7.94	8.10	0.052	0.582	1.71	0.755	0.88	2.90	+0.37
UGC04325	8	2.45	0.409	0.126	6.45	6.61	0.049	0.500	2.23	0.163	0.88	0.86	+1.53
UGC05414	6	3.72	0.930	0.555	7.72	7.30	0.137	0.181	0.97	0.414	0.83	0.84	-0.63
UGC05918	8	1.13	0.188	0.020	5.13	5.29	0.129	0.063	0.86	0.704	1.00	0.33	+1.44
UGC05999	5	0.06	0.021	0.004	4.06	3.28	0.013	0.331	1.99	0.155	1.00	0.15	+1.60 c
UGC06628	7	0.02	0.005	0.000	4.02	3.92	0.025	0.113	1.14	0.963	1.00	0.49	+1.95 c
UGC06917	11	6.63	0.737	0.325	10.63	11.43	0.086	0.175	1.63	0.693	0.82	2.32	-3.14
UGC06923	6	2.83	0.709	0.414	6.83	6.42	0.226	0.500	1.97	0.384	0.83	2.40	-0.39
UGC06930	10	0.84	0.104	0.001	4.84	5.44	0.044	0.103	0.94	0.566	1.00	0.75	+1.98 c
UGC06973	9	2.55	0.365	0.077	6.55	6.95	0.036	0.051	0.86	0.299	1.00	0.23	+1.07
UGC07261	7	0.45	0.090	0.006	4.45	4.34	0.041	0.113	0.76	0.493	1.00	1.42	+1.75 c
UGC07559	7	0.54	0.109	0.010	4.54	4.43	0.152	0.181	0.98	0.604	1.00	0.48	+1.51 c
UGC07608	8	0.94	0.157	0.012	4.94	5.10	0.092	0.223	0.97	0.562	1.00	0.51	+2.07
UGC07690	7	3.28	0.655	0.342	7.28	7.17	0.103	0.113	1.10	0.446	0.86	2.92	+1.95
UGC07866	7	1.05	0.211	0.042	5.05	4.95	0.170	0.113	0.76	0.396	1.00	0.49	+1.44
UGC10310	7	0.34	0.069	0.003	4.34	4.24	0.049	0.560	2.86	0.117	1.00	0.59	+1.70 c
UGCA281	7	1.43	0.286	0.079	5.43	5.32	0.186	0.736	1.96	0.206	1.00	0.91	+1.07
CamB	9	0.28	0.039	0.000	4.28	4.67	0.044	0.500	2.37	0.002	1.00	0.74	+2.19 c
F563-V1	6	0.55	0.137	0.031	4.55	4.13	0.110	0.240	1.38	0.775	1.00	0.37	+1.79
F574-2	5	0.10	0.033	0.008	4.10	3.32	0.143	0.331	1.54	0.290	1.00	0.24	+1.55 c
NGC4068	6	0.33	0.083	0.012	4.33	3.92	0.077	0.819	1.98	0.838	1.00	0.92	+1.78
NGC6789	4	1.16	0.582	0.441	5.16	3.94	0.258	0.841	1.42	0.909	1.00	1.27	+0.28
UGC02023	5	0.25	0.082	0.030	4.25	3.46	0.172	0.793	0.99	0.618	1.00	0.54	+1.44
UGC07232	4	1.41	0.707	0.507	5.41	4.19	0.186	0.841	1.79	0.876	1.00	1.17	+0.07
UGC07577	9	0.44	0.063	0.000	4.44	4.84	0.272	0.051	0.36	0.786	1.00	0.37	+1.80 c
UGC09992	5	0.01	0.004	0.000	4.01	3.23	0.018	0.960	2.26	0.991	1.00	0.29	+1.60 c

Atlas: category-A single-L fits with 1σ parameter band and weighted residuals

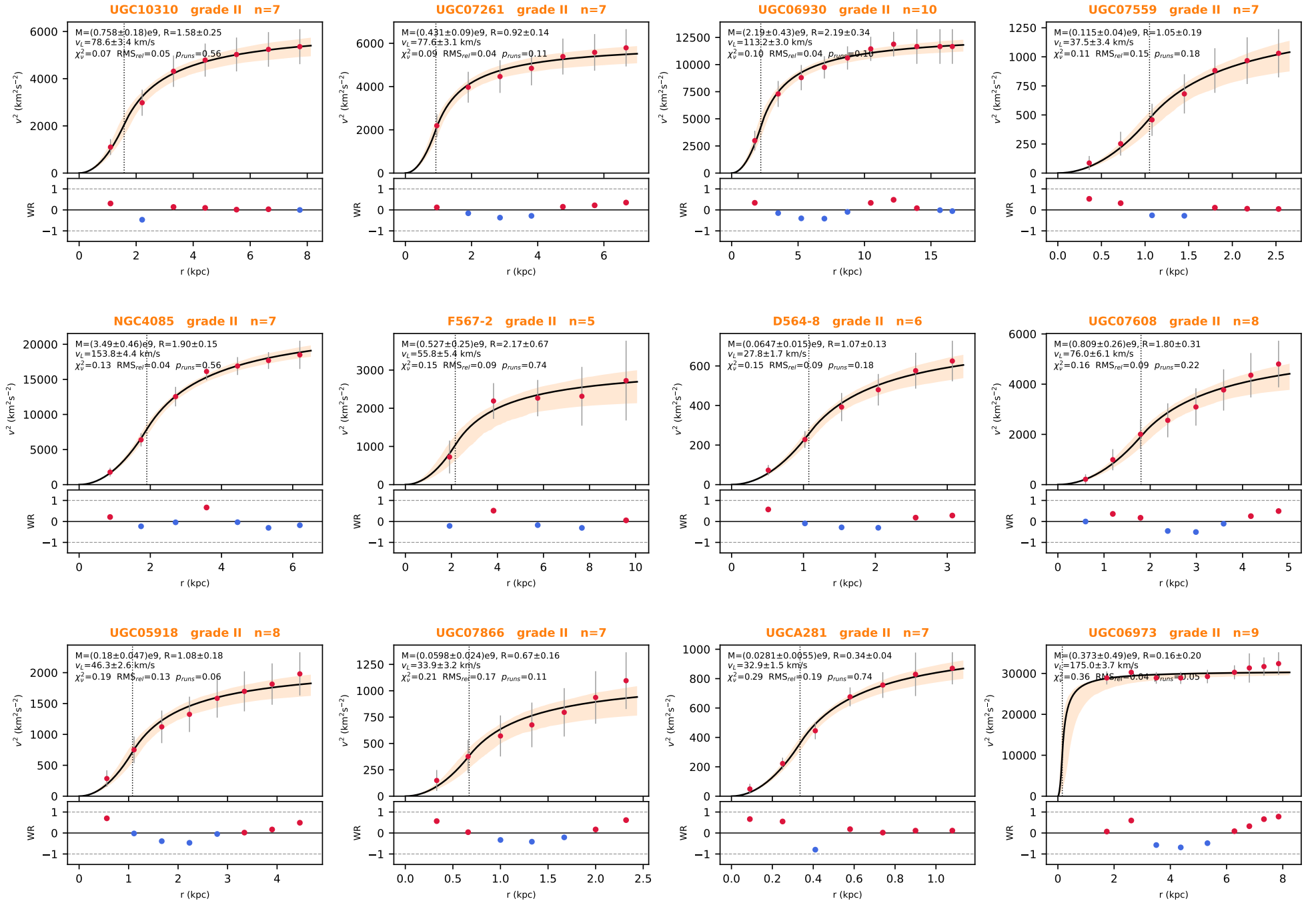
Category A single-L CL fits with 1σ parameter band and weighted residuals (sorted by grade, then χ^2_ν)



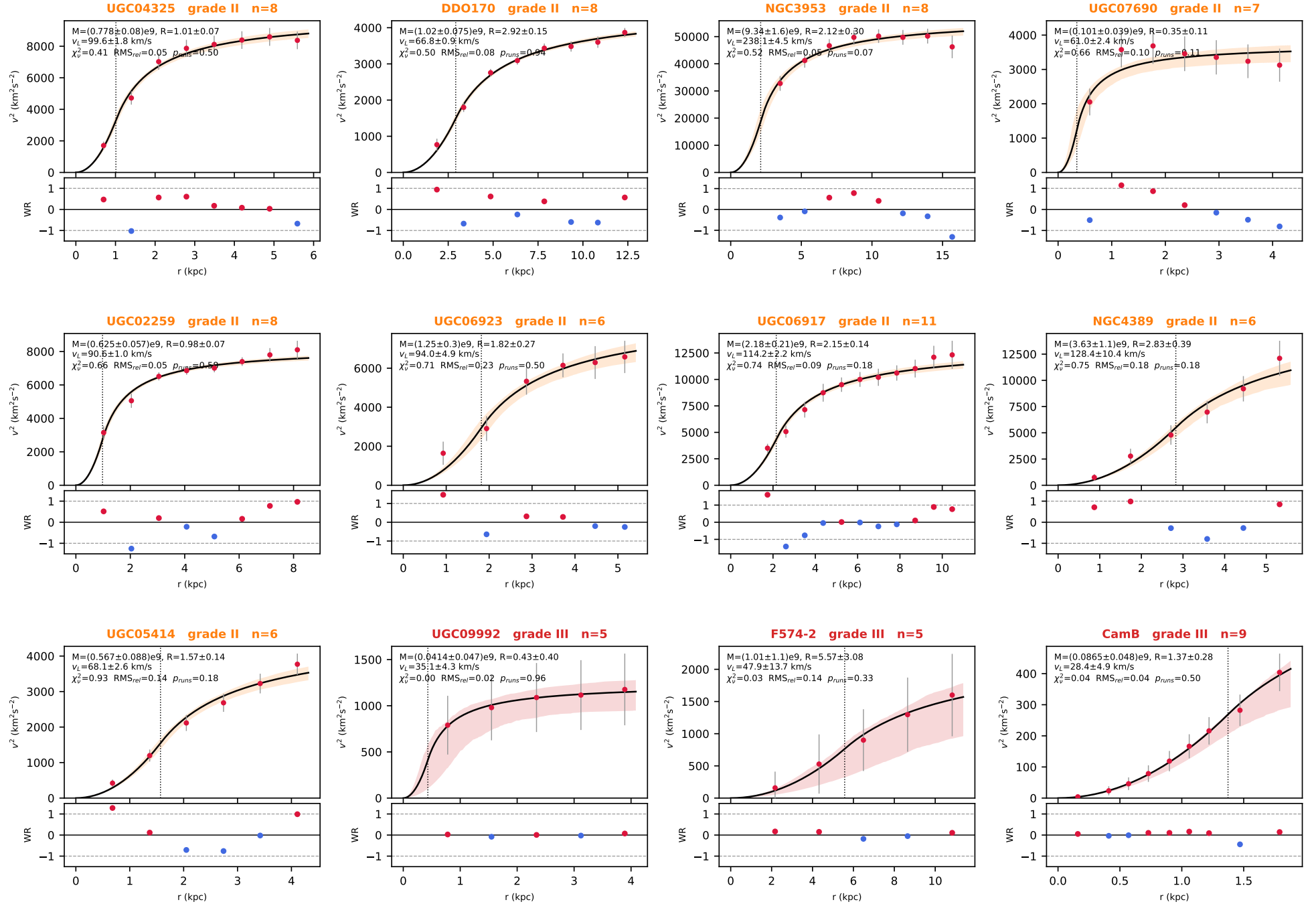
Category A single-L CL fits with 1σ parameter band and weighted residuals (sorted by grade, then χ^2_ν)



Category A single-L CL fits with 1σ parameter band and weighted residuals (sorted by grade, then χ^2_ν)



Category A single-L CL fits with 1σ parameter band and weighted residuals (sorted by grade, then χ^2_ν)



Category A single-L CL fits with 1σ parameter band and weighted residuals (sorted by grade, then χ^2_ν)

